A PROJECT PROPOSAL

ON

ASSESSMENT OF PREDISPOSING FACTORS TO TREATMENT FAILURE AMONG EXTENSIVELY DRUG-RESISTANT TUBERCULOSIS PATIENTS ATTENDING TB CLINICS IN KWARA STATE

BY

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SUMMARY

Tuberculosis (TB) is an infectious disease caused by Mycobacterium tuberculosis and remains a major public-health problem, especially in developing countries like Nigeria. Despite the availability of effective anti-TB therapy, incomplete or irregular treatment has led to the emergence of drug-resistant forms of TB, particularly extensively drug-resistant tuberculosis (XDR-TB), which poses a serious challenge to TB control programs. This study aims to determine the predisposing factors responsible for the incomplete tuberculosis treatment on the emergence of XDR-TB among patients attending selected TB Clinics. Patient treatment history will be obtained through structured questionnaires to assess adherence levels. It is expected that the study will show a higher rate of XDR-TB among patients with incomplete or interrupted TB treatment compared to those who completed therapy. The findings will provide valuable data on the relationship between treatment adherence and drug resistance, which will help strengthen TB control programs, improve patient follow-up, and guide policymakers and healthcare providers in designing effective interventions to reduce the burden of XDR-TB in Nigeria.

1.0 INTRODUCTION

Tuberculosis (TB) is a contagious and potentially life-threatening disease caused by Mycobacterium tuberculosis. It primarily affects the lungs but can also spread to other organs of the body. Despite the availability of effective anti-TB drugs, the disease remains one of the leading causes of death from infectious agents worldwide, particularly in developing countries such as Nigeria (World Health Organization, 2025). The increasing emergence of drug-resistant strains of M. tuberculosis has made its control more challenging and expensive (Centers for Disease Control and Prevention, 2023).

Extensively drug-resistant tuberculosis (XDR-TB) is a severe form of TB that arises when the bacteria become resistant to Rifampicin and Isoniazid (the two major first-line drugs), as well as to Fluoroquinolones and at least one second-line drugs (Kanamycin, Capreomycin). (WHO, 2023). One of the major causes of this resistance is incomplete or inadequate treatment of TB. When patients discontinue therapy prematurely or fail to adhere to their medication schedules, the bacteria are exposed to sub-therapeutic drug concentrations, which encourages the selection and multiplication of resistant strains (Lungu et al., 2022).

This situation poses a serious public-health threat, as XDR-TB requires longer and more toxic treatment regimens, has higher treatment failure rates, and increases the risk of mortality (Adepoju et al., 2021). Additional contributing factors include HIV co-infection, poor drug quality, lifestyle and adverse socioeconomic conditions (as seen in appendix I). Understanding how incomplete treatment contributes to the emergence of XDR-TB is therefore essential to improving adherence, strengthening TB control programs, and guiding appropriate policy interventions. This study seeks to determine the effect of incomplete tuberculosis treatment on the development of XDR-TB and to provide evidence-based data for better disease management and control.

1.1 STATEMENT OF PROBLEM

Tuberculosis (TB) remains a major public-health challenge, particularly in developing countries like Nigeria, where it continues to cause significant illness and death despite the availability of effective treatment. The increasing emergence of drug-resistant forms of TB, especially extensively drug-resistant tuberculosis (XDR-TB), has made disease control more difficult. One of the major factors responsible for this problem is incomplete or irregular TB treatment. Many patients fail to complete their prescribed therapy due to poor adherence, side effects of medication, financial difficulties, or inadequate follow-up. When treatment is interrupted, the Mycobacterium tuberculosis organisms are exposed to sub-therapeutic drug concentrations, leading to the selection and multiplication of resistant strains. These resistant strains are more difficult and costly to treat, often resulting in prolonged illness, poor treatment outcomes, and increased risk of death.

1.2 JUSTIFICATION OF STUDY

This study is justified because incomplete tuberculosis treatment remains a major factor contributing to the emergence of extensively drug-resistant tuberculosis (XDR-TB). Poor adherence and premature discontinuation of anti-tuberculosis therapy increase the risk of treatment failure and promote the development and transmission of drug-resistant Mycobacterium tuberculosis strains. Understanding this relationship is crucial for improving patient adherence, preventing treatment failure, and limiting the spread of resistant TB. The findings from this study will provide useful data for healthcare workers and policymakers to strengthen tuberculosis control programs and promote complete treatment compliance among patients. Ultimately, this study will contribute to efforts aimed at reducing the growing burden of drug-resistant tuberculosis in Nigeria.

Connection to Sustainable Development Goals (SDGs)

This study contributes directly to SDG 3: Good Health and Well-Being, particularly target 3.3, which aims to end the epidemics of tuberculosis and other communicable diseases by 2030. By assessing factors that lead to treatment failure among XDR-TB patients, the study will provide evidence to strengthen TB control programs, improve treatment adherence, and reduce the spread of resistant TB strains.

1.3 AIM

To determine the risk factor responsible for the emergence of extensively drug resistant tuberculosis (XDR-TB).

1.4 OBJECTIVES

1. To determine the prevalence of extensively drug-resistant tuberculosis (XDR-TB) among previously treated tuberculosis patients.
2. To identify predisposing factors responsible for incomplete or interrupted anti-tuberculosis treatment.
3. To find the association between the factors responsible for the incomplete anti-tuberculosis treatment and emergence of XDR TB.

1.5 HYPOTHESIS

NULL HYPOTHESIS (H_0): There is no significant association between selected risk factors and the emergence of extensively drug-resistant tuberculosis (XDR-TB) among tuberculosis patients.

ALTERNATIVE HYPOTHESIS (H_1): There is a significant association between selected risk factors and the emergence of extensively drug-resistant tuberculosis (XDR-TB) among tuberculosis patients.

2.0 METHODOLOGY

2.1 STUDY DESIGN : This study will adopt a descriptive cross-sectional laboratory based design.

2.2 STUDY DURATION : The study will last for three months encompassing data retrieval, analysis and report writing.

2.3 STUDY POPULATION: Patients on tuberculosis treatment attending the TB Clinics in Kwara State,Ilorin will be recruited.

2.4 SAMPLE SIZE DETERMINATION : Based on an expected prevalence rate of 19.18% XDR -TB in North Central, Nigeria (Oladimeji et al., 2022), the minimum sample size would be approximately 238 participants.

2.5 STUDY AREA: The study will be conducted at Tuberculosis Diagnostic Centers in Kwara State, Ilorin.

2.6 INCLUSION CRITERIA :

- Adult patients (≥15 years) with confirmed MDR-TB, RR-TB, or XDR-TB diagnosed by molecular or culture-based methods.
- Patients with complete treatment history and laboratory records available in TB registers or hospital records.
- Patients that are willing to participate.

2.7 EXCLUSION CRITERIA :

- Pediatric patients (<15 years)
- Patients with no or incomplete treatment history.
- Patients that are not willing to participate.

2.8 ETHICAL CONSIDERATION: Ethical approval for this study will be obtained from the Health Research Ethics Committee of Kwara State University Teaching Hospital (KWASUTH), Ilorin.

2.9 DATA COLLECTION: A structured data extraction form will be used to retrieve relevant sociodemographic, clinical, and laboratory information from eligible patient records and patients will be interview for additional information.

DATA ANALYSIS

Data generated from the GeneXpert results and patient treatment histories will be analyzed using Microsoft Excel and Statistical Package for Social Sciences (SPSS) version 26.0. Descriptive statistics such as frequencies and percentages will be used to determine the proportion of XDR-TB among incompletely treated tuberculosis patients. The relationship between incomplete treatment and the occurrence of drug resistance will be assessed using the Chi-square test at a 95% confidence interval and p-value ≤ 0.05 considered statistically significant. Results will be presented in tables, charts, and graphs for easy interpretation.

EXPECTED OUTCOME

It is expected that this study will reveal the proportion of extensively drug-resistant tuberculosis (XDR-TB) among patients with incomplete tuberculosis treatment and establish a clear association between incomplete therapy and the development of drug resistance. The findings will provide valuable insight into the importance of treatment adherence and help in strengthening tuberculosis control strategies. It is also anticipated that the study outcomes will guide healthcare professionals and policymakers in implementing more effective monitoring and follow-up systems to reduce the burden of drug-resistant TB in Nigeria.

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PROPOSED BUDGET FOR THE STUDY

S/N	ITEMS	AMOUNT ₦ : K
1	Administrative & Ethical Approval	20,000:00
2	Printing of Questionnaires (250-300 copies)	10,000:00
3	Stationery(Files,Pen, Notebook etc)	5,000:00
4	Statistical Consultation	8,000:00
5	Follow-up Visits & Logistics	30,000:00
6	Phone Calls to Patients	7,000:00
7	Internet/Data for Research & Analysis	7,000:00
8	Typing & Data Entry Assistance	5,000:00
9	Spiral Binding (Draft Copies)	4,000:00
10	Final Hard Binding (3–4 Copies)	4,000:00
	TOTAL	**100,000:00**

APPENDIX I

Questionnaire on Factors Associated with Emergence of XDR-TB in Kwara State

Instructions:

This questionnaire is for research purposes only.

All answers will be kept confidential.

Tick (✓) or write your answer as appropriate.

PHONE NUMBER: _____

SECTION A: Socio-Demographic & Socio-Economic Characteristics

1. Age: _____ years **2. Sex: ☐ Male ☐ Female**

3.Marital Status: ☐ Single ☐ Married ☐ Divorced ☐ Widowed

4.Highest Education Level: ☐ No formal education ☐ Primary ☐ Secondary ☐ Tertiary

5.Occupation: ☐ Employed (formal) ☐ Employed (informal) ☐ Unemployed ☐ Student ☐ Retired

6.Monthly Income: ☐ < ₦20,000 ☐ ₦20,000–₦50,000 ☐ ₦50,000–₦100,000 ☐ > ₦100,000

7.Household Size (number of people living together): ☐ 1–3 ☐ 4–6 ☐ ≥7

8.Type of Residence: ☐ Urban ☐ Rural

9.Housing Type: ☐ Owned ☐ Rented ☐ Living with family/friends

10.Access to Clean Water: ☐ Yes ☐ No **11. Access to Toilet Facility: ☐ Yes ☐ No**

SECTION B: Clinical History

12. Have you been treated for TB before? ☐ Yes ☐ No **13. If yes, how many times? ☐ Once ☐ Twice ☐ More than twice**

14.Duration of previous TB treatment: ☐ < 6 months ☐ 6–12 months ☐ > 12 months

15.Do you know your HIV status? ☐ Positive ☐ Negative ☐ Unknown

16.Do you have any other chronic illness? ☐ Yes ☐ No **17. If yes, ☐ Diabetes mellitus ☐ Advanced lung disease ☐ Others: _____**

SECTION C: Treatment Adherence & Drug Use

18. Are you currently taking TB medication as prescribed? ☐ Yes ☐ No **19. If no, ☐ Convenient time ☐ Irregular timing**

20. How do you take your prescribed TB medication dose? ☐ Full dose as prescribed ☐Sometimes reduce the dose ☐Sometimes skip doses ☐Not sure of correct dose

21. Have you ever missed TB medication for more than 2 consecutive days? ☐ Yes ☐ No

22. If yes, reason: ☐ Side effects ☐ Forgot ☐ Financial problems ☐ Felt better ☐ Drug unavailable ☐ Others: _____

23. Have you ever stopped TB treatment before completion? ☐ Yes ☐ No **24. If yes, reason: ☐ Side effects ☐ Felt better ☐Financial constraints ☐ Lack of access to drugs ☐ Others:_____**

25. Were you placed on Directly Observed Therapy (DOT)? ☐ Yes ☐ No

26. Did you ever change TB treatment facility during treatment? ☐ Yes ☐ No **27. If yes, reason: ☐ Relocation ☐ Distance to facility ☐ Poor service ☐ Personal preference ☐ Others: _____**

28. Have you ever used herbal or alternative medicine for TB? ☐ Yes ☐ No

SECTION D: Lifestyle & Risk Factors

29. Do you smoke? ☐ Yes ☐ No. **30. If yes, how often: ☐ Daily ☐ Occasionally ☐ Never**

31. Do you consume alcohol? ☐ Yes ☐ No **32. If yes, how often: ☐ Daily ☐ Occasionally ☐ Never**

33. Have you been in close contact with a drug-resistant TB patient? ☐ Yes ☐ No

34. Have you ever been incarcerated? ☐ Yes ☐ No

SECTION E: Household & Environmental Factors

35. Is your living environment overcrowded? ☐ Yes ☐ No

36. Has anyone in your household had TB before? ☐ Yes ☐ No

SECTION F: Health System & Access Factors

37. Distance to treatment center: ☐ far ☐ near

38. Have you experienced delay in receiving medication? ☐ Yes ☐ No

39. Have you ever experienced stock-out of TB drugs? ☐ Yes ☐ No

40. Can you afford to buy prescribed medications if not available at the hospital? ☐ Yes ☐ No

41. Do you have health insurance or access to free care? ☐ Yes ☐ No

Outcome Variable (from records)

XDR-TB status: ☐ Yes ☐ No